

Employing scrambled alpha-numeric randomization and RSA algorithm to ensure enhanced encryption in electronic medical records

Victor Chukwudi Osamor^{*}, Imuetinyan Boma Edosomwan

Department of Computer and Information Sciences, College of Science and Technology (CST), Covenant University, Ota, Ogun State, Nigeria

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ABSTRACT

Diverse business sectors want to ensure that their data is secure and remains confidential. A major solution to the problem of data insecurity is the concept of cryptography. Various cryptographic algorithms have been developed over the years in order to ensure maximum level of message security during transmission over insecure mediums. There is no doubt that these algorithms have been able to remarkably curb the issue of data insecurity and reduce cyber-attacks. Both asymmetric cryptographic algorithms and symmetric cryptographic algorithms are secure. However for the purpose of this paper emphasis will be laid on asymmetric cryptography. Hence, this paper emphasizes on popular asymmetric algorithms like the RSA Cryptographic algorithm. In this paper we propose a secure means of data encryption and decryption by applying the concept of scrambled alpha-numeric randomization to provide a clear understanding of the operational mechanism of the RSA algorithm during the encryption and decryption process. The scrambled alpha-numeric randomization technique employs a unique numbering and or lettering sequence to every letter of the alphabet in the case of every new message receiver. This approach also implements the cryptographic message syntax as well as the use of American standard code for information interchange (ASCII) encoding. Furthermore, the understanding of this approach will encourage new ideas for further advancements and improvement of the existing RSA cryptographic algorithm. However, the major advantage will be to increase the security of data and ensure proper authentication of data.

1. Introduction

Threats and information vulnerabilities are continuously on the increase with the development of technology. Hence the need to constantly devise new means to ensure data security. Cryptography has proven to be a trusted means to ensure privacy and security. Cryptography has been applied to various sectors one of such sector is the health sector as patients want to be rest assured that their health data is secure and the data sharing process is without compromise or interference of any sort as in Agrawal and Khan [1]. The traditional data sharing process between a medical personnel and patient is insecure and very susceptible to attacks. Figs. 1 and 2.

In the above scenario, it is very possible for unauthorized users to gain access to the health data that is expected to be kept confidential. Hence, the urgent need to implement a secure method of data sharing. [1] stressed that health information can be protected in two ways namely protection by policy which involves government or presiding bodies putting standards, rules in place to monitor and check use and transfer of health data. The other way is protection by cryptography. This paper considers exploring the second

option which is the protection by cryptography. Health information protection by cryptography will ensure that both the sender and receiver of health data have complete access and control over their data. In this paper the authors have concentrated on ensuring data authentication that is putting security measures in place to help identify and verify the authentication of the message receiver. We have proposed a scrambled alpha-numeric randomization technique which will be employed alongside the already existing RSA cryptographic algorithm. This method will help both the message sender and message receiver to encode and decode their own unique scrambling sequence as well as secure data interpretation methods.

1.1. Related work

This section aims to describe all the concepts in clear terms and describe past research work done in the area of cryptography and RSA cryptographic algorithm as a concept. Cryptography has been described as the science of secret writing [2]. Zhang et al. [3] described encryption as the process which involves transforming clear text to an unreadable format so individuals with malicious intent would have no access to it. Cryptography begins with the plain text (P) and with the use of a key (K)

^{*} Corresponding author.

E-mail addresses: vcosamor@gmail.com, victor.osamor@covenantuniversity.edu.ng (V.C. Osamor), boma.edosomwan@gmail.com (I.B. Edosomwan).

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Abbreviations	
C	Cipher Text
M	Alphabetic representation from coding sequence
E	Encryption key
D	Decryption Key
ASCII	American Standard Code for Information Interchange
CMS	Cryptographic Message Syntax
AES	Advanced Encryption Standard
DES	Data Encryption Standard
CPU	Central Processing Unit
N	RSA modulus
RSA	Rivest–Shamir–Adleman (RSA) Cryptographic Algorithm
CRT	Chinese Remainder Theorem
IBE	Identity Based Encryption

$p=223$
$q=277$
$n=61771$
$l=61272$
$e=5$
$d=85781$
Private Key $(e,n)=(5, 61771)$
Public Key $(d,n)=(85781, 61771)$

Fig. 2. Derived values from RSA algorithm.

and an encryption method (e) the plain text is encoded into a cipher text (c) [2]. Cryptographic algorithms are mainly classified into two broad umbrellas which are the symmetric cryptographic algorithms and the asymmetric cryptographic algorithms. One main significant difference between symmetric and asymmetric cryptographic algorithms is the fact that in an asymmetric cryptography system the encryption and decryption processes are done with two distinct keys referred to as the public and private keys respectively [2]. The difference may seem small but makes significant difference in ensuring the security of data. Although symmetric cryptographic algorithms are seen as faster, asymmetric is more secure and reliable [4].

Punita & Rajni [5] emphasized on the fact that some properties or conditions need to be satisfied in an asymmetric cryptography system.

I. The key generation procedure should be efficient in terms of computing.

II. Using the public and private key, both the sender and the receiver should be able to encrypt and decode any message.

III. The private key cannot be derived or computed from the associated public key.

IV. It is impossible to compute the plaintext from the public key and cipher text.

RSA being the most widely used asymmetric cryptographic algorithm ensures the private key/decryption key is kept secret while the public key is distributed for everyone involved [5]. RSA is a public key cryptographic algorithm, known as asymmetric cryptography. It uses a key pair to complete the cryptographic process. The public key is used for encryption while the private key is used for decryption. Messages encrypted with the public key can only be decrypted in a reasonable amount of time using the matching private key pair. Modulus and exponent operations are performed to generate public and private key [6]. Over the years, researchers have looked for more improved means

to make RSA safer and a lot more secure.

Nikita & Dharmendra [7]; propose a scheme for RSA cryptosystem where instead of the regular two prime numbers used for key derivation, three are used in its place. They also explored the issue of speed and looked to improve speed. Hence, the Chinese remainder theorem (CRT) is used on the RSA decryption side to improve decryption speed. Their algorithm helped to improve the security of RSA algorithm by providing a defense from some possible attacks that are common to the regular RSA algorithm. Attacks such as the common modulus attack, the chosen cipher text attack, timing attack and known plain text attack.

Amare & Vuda [8] proposed an improvement on the regular RSA algorithm such that two public keys are generated during the key generation process and these two public keys are used simultaneously instead of one as against the regular RSA cryptographic algorithm. In this scenario the public key is sent separately twice as against the traditional RSA algorithm where the public key is sent once. This makes the attacker ignorant about the key being used for encryption and thus is unable to decrypt the message. In some cases if the attacker with malicious intentions is able to intercept the sending process of both public keys, the attacker can use both public keys to decipher the encoded message. Mahajan & Singh [9] proposed a technique for data encryption where images are used. Hence, generating a unique encrypted file every time it is used to encrypt the very same message. The RSA algorithm was explored in the work of Easton [10] giving more clarification to the concept of encryption algorithm.

Meneses et al. [11] proposed to optimize the traditional RSA cryptographic algorithm by increasing the use of computational resources (Client and Server). They aim to lower time, memory, processor and network performance than all other RSA solutions. Meenal & Manoj [12] presents an RSA enabled identity based encryption (IBE) scheme that overcomes the security problems of RSA and has linear computations involved which makes key management and revocation relatively easy. Their algorithm combines IBE and RSA since IBE is a good solution to the key distribution, certificate issuing and revocation problems of public key cryptography system. They also proposed a simple mapping

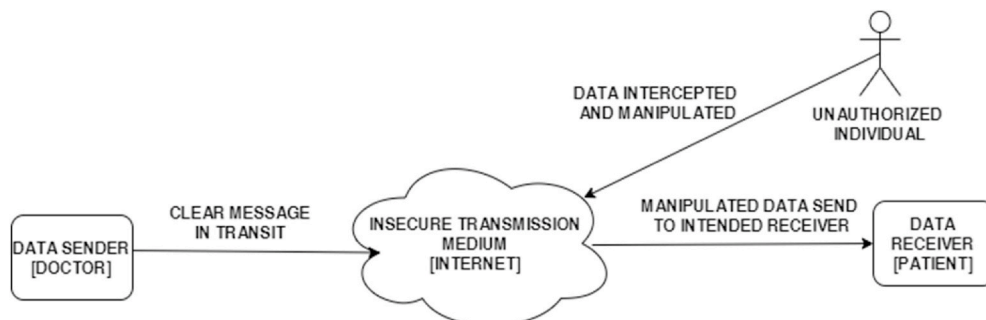


Fig. 1. Insecure data sharing scenario. Retrieved from Agrawal et al., [1].

of user ID to corresponding set of keys.

1.2. Brief overview of the cryptographic message syntax

The IETF standard for cryptographically protected messages is the Cryptographic Message Syntax. Through cryptographic techniques and protocols, it can be used to digitally sign, digest, authenticate, or encrypt any type of digital data.

Padding is a notion utilized in a block cipher where padding bytes are used to fill up the blocks. Advanced Encryption Standard (AES) employs 128-bits (16 bytes), and Data Encryption Standard (DES) uses 64-bit blocks (8 bytes). The Cryptographic Message Syntax is a padding method.

The CMS (Cryptographic Message Syntax) padding value is identical as the padding bytes number. For example for AES which uses 128 bits it will be $16 * 8$ bit characters where as for the DES which uses 512 bit key size or more it will be represented as $64 * 8$ bit characters.

Hence, in this paper the $64 * 8$ bit character block padding is employed to pad and further pre-process the raw data.

2. RSA methodology

This section describes the working of the traditional RSA algorithm and the step by step process used for key generation. This section also presents an example to illustrate the algorithm. However, for the purpose of this example in this paper relatively small prime numbers will be used for P and Q as against the large numbers used in actual RSA cryptographic implementations.

2.1. Working of RSA algorithm

RSA like every other asymmetric cryptographic algorithm aims to determine the public key/encryption key (e) as well as the private key/decryption key (d).

In RSA cryptographic algorithm the public key (e) is kept public and shared while the private key (d) as the name implies is kept hidden and secret. If Ben wants to send a message to Helen, Ben would use Helen's public key and ensure the message (m) is sent in the cipher text $c = m^e \text{ MOD } (n)$ to Helen. Then after Helen receives the cipher text from Ben she can decrypt the text by $m = c^d \text{ MOD } (n)$. The security of this algorithm largely depends on the size of the public and private keys used. They have to be significantly large figures to ensure data is not compromised in any way. The message (m) has to be secured by any means necessary.

2.2. Step by step key generation algorithm of RSA

The following steps are used to derive both the encryption key and decryption key in RSA.

1. Choose random large prime numbers for p and q.
2. Calculate RSA modulus by applying the formula: $n = p * q$.
3. Calculate Euler's Totient function $\phi(n)$ using $n = (p-1) * (q-1)$.
4. Choose a random encryption exponent value e, e must be smaller than Euler's totient function and is co-prime to $\phi(n)$. Hence $e < \phi(n)$.
5. Calculate D using the following equation: $DE = 1 \text{ mod } \phi(n)$.

2.3. RSA sample solution

Step 1. Enter two prime numbers below (p, q).

$$p = 223$$

$$q = 277.$$

Step 2. Calculate the Product of the two prime numbers using.

$$N = P * Q.$$

$$N = 223 * 277.$$

$$N = 61771.$$

Step 3. Note: L can be known as $\phi(n)$.

Calculate the value of L using:

$$L = (p - 1) * (q - 1).$$

$$L = (223-1) * (277-1).$$

$$L = 222 * 276.$$

$$L = 61272.$$

Step 4. Determining the value of E (encryption key).

Find a number between 1 and L that is coprime with L and N. Possible encryption keys are: 5,7,11,13,17,19.

Hence, E = 5.

Step 5. Determining the value of D (Decryption key).

$$D * E \text{ mod } L = 1.$$

$$85781 * 5 \text{ mod } 61272 = 1.$$

Remainder of the product of D and E when divided by L should be 1 ($D * E \% L = 1$) Possible decryption key is: 85781.

3. Concept of scrambled alpha-numeric randomization

The data security extension explained in this paper employs a scrambled alpha-numeric randomization sequence for each individual or destination. Diverse types of alpha-numeric sequence can be employed. In the example illustrated in this paper the randomized lettering sequence is used for a particular message receiver in the case of another receiver it would be a different random numbering or lettering sequence counting in 3's or 2's or even employing a mathematical counting sequence for example from the first alphabet we can have 3, 4, 6, 9, 13, 18 This technique focuses on the encryption and decryption of the message to be transmitted. It provides an extra level of security for messages in transit as it ensures data verification and authentication. However, in this technique the alpha-numeric encoding sequence has to be kept secret like the private key/decryption key as if this is shared can compromise the integrity of a message and anyone can easily decipher the contents of a message. Such additions and extension of the RSA algorithm could be applied for real time systems including Electronic Medical Records (EMR) which we will be using to explain this technique.

3.1. Encryption process using the scrambled alpha-numeric randomization technique

For the purpose of this paper the raw message before finally being encrypted with RSA encryption formula goes through some levels of message pre-processing and padding. The steps involved are:

Step 1. Scramble the message using the unique alphanumeric scrambling sequence

Step 2. Use padding method Cryptographic Message System and pad the message.

Step 3. After padding the message is then converted to ASCII code

Step 4. RSA encryption formula is applied on the ASCII code of each character.

Step 5. Final Encrypted message sent to receiver.

3.2. Decryption process using the scrambled alpha-numeric randomization technique

The following steps are involved when decrypting data using the proposed alpha-numeric scrambled randomization technique.

Step 1. RSA decryption formula is applied on the encrypted message to decode the ASCII code of each character.

Step 2. ASCII code is then decoded to reveal the Cryptographic Message Syntax interpretation which was originally encrypted.

Step 3. CMS interpretation is then converted back to scrambled text.

Step 4. The Scrambled text is decoded using the alpha numeric scrambling sequence which was applied in the encryption process and clear message is revealed.

The flow chart which illustrates the encryption and decryption process is represented below. [Figs. 3 and 4](#).

3.3. A worked example of the proposed alpha-numeric scrambled randomization technique

This section presents an example of the scrambled alpha-numeric randomization technique. We will be using the health sector to explain further.

Encryption process.

Doctor wants to send a message to one of his patients.

Patient Name: "Jonathan Henry".

Alpha-numeric sequence agreed on by both parties (Doctor and unique Patient): Random lettering sequence.

"HELLO THIS IS YOUR DOCTOR GEORGE GIVE ME A CALL LATER TODAY".

Step 1. Scramble message using the alphanumeric scrambling sequence represented (See [Tables 1 and 2](#))

Using the above alphanumeric coding sequence agreed on by the parties involved which is the Random lettering we have below the representation as:

“uqaav pulx lx evik fvwprk oqvkoq olng rq s wsaa aspqk pvfse”.

Step 2. Use padding method Cryptographic Message Syntax and pad the message

After padding (CMS):

757161617670756c786c786576696b66767770766b6f71766b6-
f716f6c6e71727173 77736161617370716b7076667365101010101
0101010101010101010.

Step 3. The message is then converted to ASCII code

55,53,55,49,54,49,54,49,55,54-55,48,55,53,54,99,55,56-54,99,
55,56-54,53,55,54,54,57,54,98-54,54,55,54,55,55,55,48,55,54,54,
98-54,102,55,49,55,54,54,98,54,102,55,49-54,102,54,99,54,101,55,
49-55,50,55,49-55,51-55,55,55,51,54,49,54,49-54,49,55,51,55,48,
55,49,54,98 55,48,55,54,54,54,55,51,54,53-49,48,49,48,49,48,49,48,
49,48,49,48,49,48,49,48,49,48,49,48,49,48,49,48,49,48,49,48,
49,48.

Step 4. RSA encryption formula is applied on ASCII code of each character. Using the already obtained values from above for e , and n above. (See Fig. 2).

$$M^e \bmod n.$$

Where:

m: Message

e: Encryption key

n: RSA modulus.

Hence, encrypted message is:

36038,5823,36038,58237,21181,58237,21181,58237,36038,21181,
36038,60364,36038,5823,21181,7965,36038,43311,21181,7965,
36038,43311,21181,5823,36038,21181,21181,42517,21181,10454,
21181,21181,36038,21181,36038,36038,36038,60364,36038,21181,
21181,10454,21181,44805,36038,58237,36038,21181,21181,10454,
21181,44805,36038,58237,21181,44805,21181,7965,21181,11935,
36038,58237,36038,511,36038,58237,36038,34216,36038,36038,
36038,34216,21181,58237,21181,58237,21181,58237,36038,34216,
36038,60364,36038,58237,21181,10454,36038,60364,36038,21181,
21181,21181,36038,34216,21181,5823,58237,60364,58237,60364,
58237,60364,58237,60364,58237,60364,58237,60364,58237,60364,
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58237,60364,58237,60364,58237,60364,58237,60364.

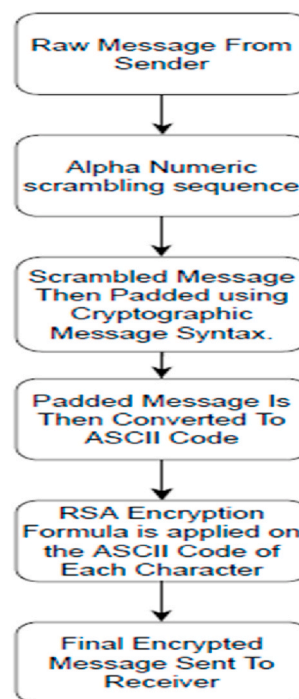


Fig. 3. Proposed encryption process.

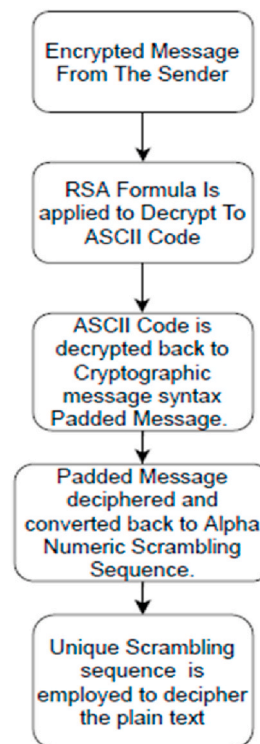


Fig. 4. Proposed decryption process.

Table 1
Alpha-numeric scrambling table.

A	B	C	D	E	F	G	H	I	J	K	L	M
S	J	W	F	Q	Y	O	U	L	C	T	A	R
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	V	Z	B	K	X	P	I	N	M	H	E	G

Table 2

Comparison of Regular RSA Crypto system and proposed Alpha numeric scrambling crypto system based on RSA possible attacks.

S/ N	Possible RSA attacks	Explanation of attack	Regular RSA cryptographic systems	Proposed RSA system
1	Factorization Attack	It's an attack that takes use of RSA's mathematical flaw, in which the attacker tries to deduce the values of p and q from the modulus in order to deduce the value of D, which is intended to be kept private.	If the attack is successful and the values of p and q are obtained the attacker can have direct access to the plain text.	If the factorization attack is successful and the value of p and q are obtained the attacker can successfully decrypt the message but will get a multiple pre-processed and encoded message as the plain text. Hence, making it hard for him/her to understand the decrypted text or message.
2	Chosen cipher text Attack	In this form of attack, the attacker selects a section of the cipher text and attempts to decrypt the plain text using the decryption exponent obtained from the cipher text. This attack can be generally applicable to public key crypto systems.	If the attack is successful and the hacker can successfully decrypt the chosen cipher text, he automatically knows the private key d and can intercept the rest of the message	If the chosen cipher text attack is successful and the attacker has successfully decrypted part of the encrypted message. The result or plain text obtained is a pre-processed and encoded message. Hence, making it difficult for him to understand.
3	Encryption exponent Attack	To shorten the encryption time, a tiny exponent (e) is employed, and the most frequent value of (e) is e=3. Once the attacker knows (e), decrypting the encrypted message is quite simple.	If the attack is successful and the corresponding value of e used is known to the hacker it will be relatively easy to intercept messages in transmission.	If the encryption exponent attack is successful and the value of e is obtained the attacker can successfully decrypt the message but will get a multiple pre-processed and encoded message as the plain text. Hence, making it hard for him/her to understand the decrypted text or message.
4	Decryption exponent Attack	This attack is of two types the revealed exponent attack and the low decryption exponent attack. The hacker obtains the decryption value of (d) via the revealed exponent attack, and once obtained, the message is easily decoded. In the second instance, because a short decryption exponent (d) or private key is utilized in the low decryption exponent attack, the hacker will have an easy time deciphering the message.	If the attack is successful and the corresponding value of d used is known to the hacker it will be relatively easy to decrypt all messages in transmission.	If the decryption exponent attack is successful and the value of d is obtained the attacker can successfully decrypt the message but will get a multiple pre-processed and encoded message as the plain text. Hence, making it hard for him/her to understand the decrypted text or message.
5	Plaintext Attack	The hacker already knows something about the plaintext in this attack, and the plaintext's properties may allow some attacks on it. Ex short messages and unconcealed messages.	If the attacker is able to successfully decipher the cipher text back to the plain message he can easily access the entire plain message	Since the attacker already knows part of the plain message it will be difficult for the hacker to successfully obtain the remaining part of the message as the message before RSA encryption has gone through processes of pre-processing padding and an extra level of pre-processing.
6	Modulus Attack	The most prevalent type is a common modulus attack, which occurs when common modulus is used (n). For instance, various users can utilize a trusted party to select the values of p and q to calculate (n), and the calculated value of (n) is used across the board for all calculations.	If the value of (n) is decoded successfully the attacker can use it to factorize the values of p and q and hence, the message is no longer secure.	If the value of (n) is known and the encrypted message is successfully decrypted by the hacker it will be difficult to understand as the message gotten as the plain text is a multiple pre-processed and encoded message.
7	Man in the middle Attack	A man in the middle (MITM) attack occurs when an attacker inserts himself in the middle of a dialogue between a user and an application or another user to eavesdrop or impersonate one of the parties, making it appear as though a regular exchange of information is taking place.	If the man in the middle is able to intercept information and is able to get the encryption key (e,n) as well as the decryption key (d,n) used. The messages can easily be intercepted.	The various levels of message pre-processing and message padding provide an extra level of security.
8	Brute force Attack	The hacker utilizes a trial-and-error method to estimate the encryption and decryption keys, as well as the values of p and q, in this attack.	RSA can be easily breakable if the values of p and q are known using the brute force algorithm. Hence, the third party will be able to factor the values of p and q in no time.	If the brute force attack is successful and the value of p and q are obtained the attacker can successfully decrypt the message but will get a multiple pre-processed and encoded message as the plain text. Hence, making it hard for him/her to understand the decrypted text or message.
9	Statistical Attack	It's a type of cyber-attack that takes advantage of statistical flaws in a certain program. For example, if an attacker understands how a random number generator works, he or she can use that information to crack an algorithm.	If the attacker is able to understand how the random number generation system works. The crypto system can be exploited.	The various levels of message pre-processing and message padding provide an extra level of security.

For a clear understanding a more concise example is shown below.

- Using the first word in the original message "HELLO"
- From the Alpha-numeric scrambling sequence table "Hello: uqaav"
- After padding using cryptographic message syntax: Hello is seen as: 7571616176
- After converting to ASCII code we have: 55,53,55,49,54,49,54,49,55,54
- Applying RSA encryption formula: $m^e \text{ MOD } n$

Where:

m: Message

e: Encryption key

n: RSA modulus.

Hence, encrypted message is:

$$55^5 \text{ MOD } 61771 = 36038.$$

$$53^5 \text{ MOD } 61771 = 5823.$$

$$55^5 \text{ MOD } 61771 = 36038.$$

$$49^5 \text{ MOD } 61771 = 58237.$$

$$54^5 \text{ MOD } 61771 = 21181.$$

$$49^5 \text{ MOD } 61771 = 58237.$$

$$54^5 \text{ MOD } 61771 = 21181.$$

$$49^5 \text{ MOD } 61771 = 58237.$$

$$55^5 \text{ MOD } 61771 = 36038.$$

$$54^5 \text{ MOD } 61771 = 21181.$$

Encrypted message is represented as: 36038,5823,36038,58237,21181,58237,21181,58237,36038,21181. Where:

- Applying RSA decryption formula: $m^d \text{ MOD } n$

m: Message
 d: Decryption key
 n: RSA modulus.
 Hence, decrypted message is:

- Decrypting: 36038, 5823, 36038, 58237,21181,58237,21181, 58237,36038,21181

$36038^{85781} \text{ MOD } 61771 = 55.$
 $5823^{85781} \text{ MOD } 61771 = 53.$
 $36038^{85781} \text{ MOD } 61771 = 55.$
 $58237^{85781} \text{ MOD } 61771 = 49.$
 $21181^{85781} \text{ MOD } 61771 = 54.$
 $58237^{85781} \text{ MOD } 61771 = 49.$
 $21181^{85781} \text{ MOD } 61771 = 54.$
 $58237^{85781} \text{ MOD } 61771 = 49.$
 $36038^{85781} \text{ MOD } 61771 = 55.$
 $21181^{85781} \text{ MOD } 61771 = 54.$

- Converting ASCII code back to originally encrypted message: 55,53,55,49,54,49,54,49,55,54 7571616176
- Converting Cryptographic Message Syntax padding value back to scrambled text: Uqav
- Converting from alpha numeric scrambled sequence back to clear text Hello.

3.4. Pros and Cons of scrambled alpha- numeric randomization technique

Advantages.

1. This technique provides more security as a third party without the alpha-numeric coding sequence employed in encryption cannot get access to clear message.
2. Vulnerability of system is reduced
3. Ensures authentication and integrity of data as only the individual with the correct alpha-numeric sequence can have access to the clear text.
4. Easy to use and simple to understand.

Disadvantages.

1. This technique does not consider the time spent for both encryption and decryption process.
2. The computation for this technique could be CPU intensive.

3.5. Evaluation of proposed system based on possible RSA attacks

The table below presents a comparison between a regular RSA crypto system and the proposed Alpha numeric scrambling sequence crypto system based on possible attacks on RSA cryptographic algorithm. (See Table 2).

4. Results and discussion

Following Osamor et al.[13] and Osamor & Osamor [14], the results are reported in separate tasks from Task 1-7. The use of similar technique even transcends to big data [14,15] with diverse application areas. TASK 1: Derive the values for p, q, n, EULERS TOTIENT, e and d using RSA algorithm.

The regular RSA algorithm was employed to enable us derive values for p, q, n, Euler's totient function, e and d. Hence, for the key generation we followed a step by step mathematical computation to derive these values as we would require these values going forward.

However, for the purpose of simplicity relatively small numbers were used for the values of p and q. in the future if this technique is to be applied large numbers should be used to ensure optimum security of

data.

TASK 2: Involved Parties Computing and Agreeing On a Unique Alpha-Numeric Coding Sequence.

Parties involved in the message transmission process (sender and receiver) need to define a specific alpha-numeric encoding sequence that will be used when encrypting and decrypting files. In the example explored in this paper we used random numbering sequence. Where the alphabets assigned to each alphabet is randomly assigned and does not follow any mathematical computation whatsoever. For the purpose of a different message receiver the alpha-numeric coding sequence has to change. It could follow a mathematical pattern as earlier described in the body of this work or could simply be randomly selected. However, in the future if this technique is employed the alpha-numeric encoding sequence agreed to employ has to be kept secret like the private key/ decryption key. If this is shared, anyone can easily decipher the contents of a message thereby exposing the privacy of its contents.

TASK 3: Encoding Data Using The Alpha-Numeric Scrambled Coding Sequence.

When the parties involved in a message sending process (sender and receiver) have derived a unique and secret alpha-numeric encoding sequence, we now move to use the alpha-numeric scrambling encoding sequence to pre-process the raw message.

TASK 4: Employing The Cryptographic Message Syntax Method To Ensure Message Is Padded According To PKCS Standards.

After the message has been successfully scrambled. The next step is to ensure that the scrambled text is padded using the cryptographic message syntax that uses 64 * 8 bit characters for data encryption standard crypto systems.

TASK 5: Encoding To ASCII Code.

After the message has been successfully padded using the cryptographic message syntax an extra layer of message pre-processing is introduced as the message is then encoding to ASCII code before it is encrypted using the RSA encryption formula.

TASK 6: Encrypting Data Using RSA Encryption Algorithm.

After the encoding to ASCII is completed. RSA encryption formula is applied on the ASCII code of each character. The RSA encryption formula $m^e \text{ MOD } n$ is employed during the encryption process.

TASK 7: Decrypting Data Using RSA Decryption Algorithm.

After the encrypted message has been successfully sent across to the receiver the RSA decryption formula ($m^d \text{ MOD } n$) is applied on the encrypted message in order to decrypt and transform back to the originally encoded ASCII code. After which the ASCII code is decoded back to the cryptographic message syntax padding form, from where it is decoded back to the scrambled text and then finally the scrambled text is decoded to get the clear text. Wang & Liu [16] used C++ class library to develop a secure RSA encryption algorithm class library. The researchers discovered groupware encapsulation with 32 bit windows platform. Our work has now extended the RSA algorithm equipped with scrambled alpha-numeric randomization technique that employs a unique numbering in hospital record management with encryption and decryption capabilities.

5. Conclusion and future work

In this paper we propose a means to ensure data authentication and data integrity by using the traditional and already existing RSA algorithm alongside the scrambled alpha-numeric randomization technique. This makes it harder for an individual with malicious intentions to manipulate data in transmission as it will be impossible to decipher the message without the alpha-numeric encoding sequence which would be kept secret like the decryption key/private key. The proposed technique is simple and easy to understand. It is used for a system that need high security of important data but with less speed for both encryption and decryption time. Encryption and decryption time is not at all considered here so in future work encryption and decryption time can be taken into consideration so the general time for the entire process is reduced. Also,

for future work large numbers can be used to measure the performance of this technique as in this paper relatively small numbers are engaged.

Authors' contributions

Victor Chukwudi Osamor, Imuetinyan Boma Edosomwan: initiated the idea of the work and both wrote the manuscript. Victor Chukwudi Osamor: supervised the work. The two authors read and approved the final manuscript.

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Declaration of competing interest

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